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RESEARCH TOPIC

Recent trends in neuromorphic computing hardware for edge AI applications

EXECUTIVE SUMMARY

Topic Framing

Neuromorphic computing is an emerging field that aims to mimic the neural architecture of the human brain to achieve efficient computation, particularly for edge AI applications. This synthesis focuses on recent advancements in neuromorphic computing hardware, emphasizing its application in edge AI, where computational efficiency and low power consumption are critical.

Most Relevant Papers and Main Contributions

1. **"Hardware, Algorithms, and Applications of the Neuromorphic Vision Sensor: A Review"** discusses the integration of neuromorphic vision sensors in edge AI applications. These sensors are designed to process visual information in a manner similar to biological systems, offering significant improvements in energy efficiency and processing speed.
2. **"Neuromorphic computing for robotic vision: algorithms to hardware advances"** highlights the role of neuromorphic computing in enhancing robotic vision systems. The paper reviews the transition from algorithms to hardware implementations, emphasizing the potential for real-time processing and reduced energy consumption.
3. **"A Review on Fundamentals, Applications, Challenges and Current Status of Spiking Automotive Electronics"** addresses the application of neuromorphic computing in automotive systems, particularly for autonomous vehicles. The review distinguishes between demonstrated implementations and speculative applications, providing a critical analysis of the efficiency claims under real driving conditions.
4. **"Hafnium-Based Ferroelectric Post-Moore Electronics: Device Physics, Integration Architectures, and Neuromorphic System Implementation"** explores the use of hafnium-based ferroelectric materials in neuromorphic systems. These materials offer compatibility with existing semiconductor processes and potential for high scalability, crucial for edge AI applications.

5. **"CMOS compatible probabilistic computing hardware with cointegrated reconfigurable p-bits and synapse arrays"** presents a novel approach to probabilistic computing using charge-trap field-effect transistors. This technology integrates probabilistic bits and synapse arrays, offering a scalable solution for neuromorphic computing at the edge.

Recurring Trends and Methodological Patterns

- **Event-Driven Processing:** A common theme across the papers is the emphasis on event-driven processing, which is central to neuromorphic computing. This approach allows systems to react to changes in the environment in real-time, significantly reducing power consumption compared to traditional methods.
- **Material Innovation:** The use of advanced materials, such as hafnium-based ferroelectrics and two-dimensional materials, is a recurring trend. These materials are crucial for developing scalable and efficient neuromorphic hardware.
- **Integration with Existing Technologies:** Many studies focus on integrating neuromorphic systems with existing semiconductor technologies, such as CMOS, to facilitate widespread adoption and compatibility with current manufacturing processes.

Limitations and Evidence Gaps

- **Real-World Validation:** While many papers propose promising technologies, there is a noted gap in real-world validation. The efficiency and performance claims often lack extensive testing under practical conditions, particularly in automotive and robotic applications.
- **Scalability Challenges:** Despite advances in material science and device architecture, scalability remains a significant challenge. The integration of new materials and technologies into large-scale production processes is not yet fully realized.
- **Speculative Applications:** Some reviews mix demonstrated applications with speculative ones, which can lead to overestimation of the current capabilities of neuromorphic systems. Clear differentiation between proven and theoretical applications is necessary for accurate assessment.

Conclusion

Recent advancements in neuromorphic computing hardware for edge AI applications show promising directions, particularly in energy efficiency and real-time processing capabilities. However, the field faces challenges in scalability, real-world validation, and distinguishing between speculative and demonstrated applications. Continued research and development are needed to address these limitations and fully realize the potential of neuromorphic computing in edge AI.

1

A Comprehensive Survey of Federated Learning for Edge AI: Recent Trends and Future Directions

EuropePMC

2025-12-02

DOI: 10.20944/preprints202512.0118.v1

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/PPR/PPR1129854>

2

Hardware, Algorithms, and Applications of the Neuromorphic Vision Sensor: A Review.

EuropePMC

2025-10-07

DOI: 10.3390/s25196208

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41095030>

3

Neuromorphic computing for robotic vision: algorithms to hardware advances.

EuropePMC

2025-08-13

DOI: 10.1038/s44172-025-00492-5

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/40804110>

4

A Review on Fundamentals, Applications, Challenges and Current Status of Spiking Automotive Electronics

OpenAlex

2026-04-16

DOI: <https://doi.org/10.30939/ijastech..1845650>

Abstract. Automotive edge systems face a growing gap between computational demand and what vehicle platforms can supply under tight power and thermal budgets, especially in autonomous vehicles. Neuromorphic computing is proposed as response, owing to its event driven operation. But earlier reviews on this subject tend to mix demonstrated uses with speculative applications and do not always relate efficiency claims to real driving conditions. This review addresses this gap in the literature; automotive system integration of neuromorphic hardware, spiking neural network training and deployment, event based sensing. Reviewed studies are separated into demonstrated implementations with measurable outcomes on stated platform and proposed opportunities that still lack automotive grade validation. Four observations are obtained from this review. First, efficiency gains from spike based processing become credible mainly when the workload is sparse and temporal by nature and when coding policy is selected with bounded time to decision in mind. Second, cross study comparison remains difficult because latency, energy, event rate condition, and stopping rule are usually reported in inconsistent ways across

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5

Hafnium-Based Ferroelectric Post-Moore Electronics: Device Physics, Integration Architectures, and Neuromorphic System Implementation

OpenAlex

2026-04-13

DOI: <https://doi.org/10.1007/s40820-026-02158-z>

Abstract. Hafnium-based ferroelectric (Hf-FEs) materials overcome the limitations of perovskite ferroelectric materials. Owing to their compatibility with the complementary metal-oxide-semiconductor process and high scalability, Hf-FEs devices have driven the development of non-volatile memory and neuromorphic computing, demonstrating significant potential for application in image processing and in-memory logic operations. First of all, this paper summarizes the material systems, device structure, and physical mechanisms relevant to Hf-FEs devices. Then, for the purpose of achieving efficient neuromorphic computing, the evaluation parameters related to Hf-FEs devices, specifically concerning storage performance and synaptic plasticity, are discussed. Furthermore, the progress of Hf-FEs devices in arrays and hardware integration is systematically reviewed, offering insights for future applications. Finally, this study explores in depth the prospects and challenges of these devices in advanced applications, providing valuable guidance for the development of high-performance neuromorphic computing devices.

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6

Tactile and Visual Artificial Synaptic Devices: Progress and Challenges

OpenAlex

2026-04-15

DOI: <https://doi.org/10.3390/electronicmat7020008>

Abstract. The von Neumann architecture faces a “memory wall” problem due to the physical separation of memory and processor, posing major challenges to energy efficiency and latency in the era of artificial intelligence. To overcome these bottlenecks, artificial synaptic devices inspired by biological systems have emerged as an important research direction. By integrating sensing and computing functions at the device level, these architectures provide a promising approach for the efficient processing of natural physical signals. Supported by advances in functional materials and artificial neural network (ANN) algorithms, artificial synaptic devices are capable of perceiving and processing various external stimuli, showing strong potential for applications in intelligent electronic skins, robotics, and edge computing. This review provides a comprehensive overview of recent advances in artificial synaptic devices, with particular emphasis on tactile and visual sensing applications. We discuss representative device types and operating mechanisms, analyze critical challenges from the perspectives of material engineering and functional integration, and further summarize potential solutions and fu

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7

CMOS compatible probabilistic computing hardware with cointegrated reconfigurable p-bits and synapse arrays

OpenAlex

2026-04-11

DOI: <https://doi.org/10.1038/s41467-026-71906-x>

Abstract. Probabilistic computing (p-computing) has emerged as a promising approach to solve combinatorial optimization problems (COPs) at room temperature, offering a practical alternative to quantum computing. Scalable p-computing hardware requires integration of probabilistic bits (p-bits) and synapses, two components with inherently contrasting device requirements. Here, we present a charge-trap field-effect transistor (CT-FET) with a silicon-oxide-nitride-silicon gate stack that functions reconfigurably as both a p-bit and a synapse, derived from a commercialized flash memory structure. Gate voltage control of trap energy enables stochastic shallow trap operation for p-bits and stable deep trap operation for synapses. P-bits and synapse array are cointegrated on a single wafer using standard CMOS processes, enabling scalable p-computing hardware. Experimental validation shows successful solution of a 4-node weighted max-cut problem including negative weights, while simulations confirm efficient solution of a 20-node weighted max-cut problem including negative weights via chaotic simulated annealing. These results demonstrate a CMOS compatible pathway toward practical and scalable probab

Source URL. <https://openalex.org/W7153600740>

8

Dedicated and Reconfigurable Artificial Neurons and Synapses based on Two-Dimensional Materials for Efficient Neuromorphic Application.

EuropePMC

2026-03-31

DOI: [10.1007/s40820-026-02139-2](https://doi.org/10.1007/s40820-026-02139-2)

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41915333>

9

What is next for LLMs? Pushing the boundaries of next-gen AI computing hardware with photonic chips.

EuropePMC

2025-10-06

DOI: [10.1515/nanoph-2025-0217](https://doi.org/10.1515/nanoph-2025-0217)

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41209406>

10

Quality of experience-aware application deployment in fog computing environments using machine learning.

EuropePMC

2025-09-05

DOI: [10.7717/peerj-cs.3143](https://doi.org/10.7717/peerj-cs.3143)

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/40989363>

11

Patient-Centric Edge Intelligence for Hospital Monitoring: Embedded Anomaly Detection, Ethical Governance and Selective Telemetry

OpenAlex

2026-04-14

DOI: <https://doi.org/10.25258/ijddt.16.11s.59>

Abstract. Modern hospitals use patient monitoring systems that generate substantial data streams and send them to the cloud, facing challenges such as data streams, costs, privacy issues, and latency that are directly linked to cloud-based analysis or architecture. Our project addresses the limitations of existing systems by enabling local inference and embedded anomaly detection with ethical governance and selective telemetry. This paper presents a patient-centric edge intelligence framework integrated into devices that feature anomaly detection, selective telemetry, and governance-aware selection. The architecture operates by embedding intelligence at the edge to detect clinically relevant anomalies locally, and by escalating complex data scenarios to the cloud only when necessary. The evaluation strategy focuses on hospital monitoring scenarios, emphasizing latency reduction, bandwidth efficiency, and privacy preservation. Our paper presents a novel illustration of how edge intelligence can enhance patient care, reduce reliance on cloud computing, and support the ethical deployment of AI in future clinical environments.

Source URL. <https://openalex.org/W7154476230>

12

Quantum spectral clustering: Comparing parameterized and neuromorphic quantum kernels

OpenAlex

2026-04-13

DOI: <https://doi.org/10.1063/5.0291037>

Abstract. We compare a parameterized quantum kernel (pQK) with a quantum leaky integrate-and-fire (QLIF) neuromorphic computing approach that employs either the Victor–Purpura or van Rossum kernel in a spectral clustering task, as well as the classical radial basis function kernel. To our knowledge, this is the first study to employ quantum neuromorphic leaky integrate-and-fire neurons as kernel generators within a spectral clustering framework and to directly compare them against gate-based quantum fidelity kernels under identical conditions. Performance evaluation includes label-based classification and clustering metrics, as well as optimal number of clusters predictions for each dataset based on an elbow-like curve, as is typically used in K-means clustering. The pQK encodes feature vectors through angle encoding, with rotation angles scaled parametrically. Parameters are optimized through grid search to maximize kernel-target alignment, producing a kernel that reflects distances in the feature space. The quantum neuromorphic approach uses population coding to transform data into spike trains, which are then processed using temporal distance metrics. Kernel matrices are used as input int

Source URL. <https://openalex.org/W4416105047>

13

Super-Constant Weight Dicke States in Constant Depth Without Fanout

arXiv

2026-04-16

Abstract. An n -qubit Dicke state of weight k , is the uniform superposition over all n -bit strings of Hamming weight k . Dicke states are an entanglement resource with important practical applications in the NISQ era and, for instance, play a central role in Decoded Quantum Interferometry (DQI). Furthermore, any symmetric state can be expressed as a superposition of Dicke states. First, we give explicit constant-depth circuits that prepare n -qubit Dicke states for all $k \leq \text{polylog}(n)$, using only multi-qubit Toffoli gates and single-qubit unitaries. This gives the first QAC^0 construction of super-constant weight Dicke states. Previous constant-depth constructions for any super-constant k required the FANOUT_n gate, while QAC^0 is only known to implement FANOUT_k for k up to $\text{polylog}(n)$. Moreover, we show that any weight- k Dicke state can be constructed with access to $\text{FANOUT}_{\min(k, n-k)}$, rather than FANOUT_n . Combined with recent hardness results, this yields a tight characterization: for $k \leq n/2$, weight- k Dicke states can be prepared in QAC^0 if and only if $\text{FANOUT}_k \in \text{QAC}^0$. We further extend our tech

Source URL. <http://arxiv.org/abs/2604.15298v1>

14

Determining institutional discrepancies for ERM software applications

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.15556/ijim.02.01.002

Abstract. No abstract available.

Source URL. <https://doi.org/10.15556/ijim.02.01.002>

15

Structural Identification of Large Finite Element Models Using Commodity Computing Clusters for Parallel Genetic Algorithms

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.12783/shm2015/112

Abstract. No abstract available.

Source URL. <https://doi.org/10.12783/shm2015/112>

16

A Birkhoff type transitivity theorem for non-separable completely metrizable spaces with applications to linear dynamics

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.7900/jot.2011may12.1971

Abstract. No abstract available.

Source URL. <https://doi.org/10.7900/jot.2011may12.1971>

17

Trends and Strategies in Implementing RME and PJBL for Teaching Central Tendency: A Systematic Literature Review

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.31851/indiktika.v7i2.20054

Abstract. Penelitian ini bertujuan untuk mengkaji tren penelitian terkini dan strategi yang paling efektif untuk mendukung kemampuan berpikir kritis siswa pada materi ukuran pemusatan data. Metode yang digunakan adalah Systematic Literature Review (SLR) dengan mengumpulkan artikel-artikel yang relevan dari tahun 2018 hingga 2025, yang mengkaji tren dan strategi dalam konteks pembelajaran ukuran pemusatan data. Artikel-artikel yang relevan dikumpulkan dari berbagai basis data akademik terkemuka, seperti Scopus dan Google Scholar, dengan menggunakan kata kunci yang berkaitan dengan Realistic Mathematics Education (RME), Project Based Learning (PjBL), dan ukuran pemusatan data. Proses seleksi inklusi dilakukan dengan menyaring 1383 artikel yang relevan menjadi 175 artikel, dan setelah pemeriksaan lebih lanjut, 147 artikel dieliminasi berdasarkan kriteria relevansi. Akhirnya, 15 artikel dipilih untuk analisis lebih lanjut. Hasil utama penelitian ini menunjukkan bahwa RME dan PjBL menjadi dua pendekatan yang semakin populer dari tahun 2018 hingga 2025 dalam pengajaran ukuran pemusatan di kalangan peneliti, dengan strategi penggunaan konteks dunia nyata, proyek berbasis pembelajaran, dan integrasi

Source URL. <https://doi.org/10.31851/indiktika.v7i2.20054>

18

APPLICATIONS OF NANOTECHNOLOGY IN FOOD PROCESSING AND PACKAGING

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.33899/magrij.2022.134238.1178

Abstract. No abstract available.

Source URL. <https://doi.org/10.33899/magrij.2022.134238.1178>

19

Evolution and Applications of Artificial Intelligence in SAls

CrossRef

This is an accepted article with a DOI pre-assigned that is not yet published.

DOI: 10.56251/yxek7242

Abstract. No abstract available.

Source URL. <https://doi.org/10.56251/yxek7242>

arXiv

2026-04-16

Abstract. We prove a strong general-purpose bound for the diameter of a finite group depending only on the diameters of its composition factors and the maximal exponent of a normal abelian section. There are a number of notable applications: (1) if G is a finite soluble group of exponent e , $\mathrm{diam}(G) \leq (\log |G|)^{e^8}$, (2) anabelian groups with bounded-rank composition factors have polylogarithmic diameter, (3) transitive soluble subgroups of S_n have diameter $\leq n^5$, and (4) Grigorchuk's gap conjecture holds for any finitely generated group acting faithfully on a bounded-degree rooted tree. Additionally, conditional on Babai's conjecture, (5) any transitive permutation group of degree n has diameter bounded by a polynomial in n (a folkloric conjecture), and (6) Grigorchuk's gap conjecture holds for residually finite groups, and thus the conjecture reduces to the simple case.

Source URL. <http://arxiv.org/abs/2604.15303v1>

OpenAlex

2026-04-16

DOI: <https://doi.org/10.1038/s41467-026-71937-4>

Abstract. We report that an autoencoder-based neuromorphic architecture, combined with Fowler-Nordheim annealing, is sufficient to implement scalable higher-order Ising machines. We show that these machines can consistently produce state-of-the-art solutions with high reliability and with competitive time-to-solution metrics. The autoencoder captures higher-order interactions by decomposing Ising clauses and Ising spins into encoder-decoder layers of spiking neurons, thereby keeping the resource complexity independent of the interaction order for sparse problems. An annealing process based on the dynamics of Fowler-Nordheim quantum mechanical tunneling extrapolates between an $O(1)$ annealing schedule and an $O(\log t)$ annealing schedule.

Source URL. <https://openalex.org/W4414987987>

EuropePMC

2025-11-10

DOI: 10.1016/j.patter.2025.101414

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41472829>

23

Triboelectric nanogenerators for neural data interpretation: bridging multi-sensing interfaces with neuromorphic and deep learning paradigms.

EuropePMC

2025-11-07

DOI: 10.3389/fncom.2025.1691017

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41281720>

24

Towards the neuromorphic Cyber-Twin: an architecture for cognitive defense in digital twin ecosystems.

EuropePMC

2025-11-04

DOI: 10.3389/fdata.2025.1659757

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/41262924>

25

Recent advances in the inverse design of silicon photonic devices and related platforms using deep generative models.

EuropePMC

2025-06-11

DOI: 10.7717/peerj-cs.2895

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/40567675>

26

Biosignal-integrated robotic systems with emerging trends in visual interfaces: A systematic review.

EuropePMC

2024-02-21

DOI: 10.1063/5.0185568

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/38510371>

27

Monitoring time domain characteristics of Parkinson's disease using 3D memristive neuromorphic system.

EuropePMC

2023-12-15

DOI: 10.3389/fncom.2023.1274575

Abstract. No abstract available.

Source URL. <https://europepmc.org/article/MED/38162516>

28

Not Only are Protectionist Measures Harmful for Economic Development but the Recent Changes in the Global Trading System have Rendered them Irrelevant

CrossRef

2014-02-05

DOI: 10.2139/ssrn.2389355

Abstract. No abstract available.

Source URL. <https://doi.org/10.2139/ssrn.2389355>

29

Integration Technologies of Carbon-Based Materials in Computing-in-Memory Chips

OpenAlex

2026-04-13

DOI: <https://doi.org/10.54254/2755-2721/2026.bj32738>

Abstract. As the traditional Von Neumann architecture computing system with separate storage and computing units no longer meets people's needs, the realization of the integrated storage and computing technology that combines the two is extremely urgent. However, due to the inherent limitations of traditional silicon-based materials, carbon-based materials, with their superior properties, have become the ideal material for realizing the integrated computing and storage technology. At present, this technology is widely applied in fields such as three-dimensional integrated circuits, sensor detection, and digital computing. This review will systematically describe the integration techniques of carbon-based materials in memory-computing integrated chips, providing a detailed analysis of their development and challenges in digital circuits, computing, and sensing applications. Furthermore, this paper explores corresponding solutions and outlines the future prospects for carbon-based compute-in-memory chips. In addition, this study has important theoretical value for promoting fundamental innovations in post-Moore era integrated circuit architectures, providing a feasible material basis and techn

Source URL. <https://openalex.org/W7154016131>

30

Gravitational-wave lensing beyond rays: a disordered-system approach

arXiv

2026-04-16

Abstract. We develop a framework to describe gravitational wave propagation through a stochastic distribution of weak gravitational lenses beyond the geometric optics limit. We model the lens distribution as a static random background field and formulate the problem in the language of quenched disorder, treating the disorder averaged density matrix as the fundamental object from which observables are computed. Using the Schwinger Keldysh formalism, we construct a path-integral representation of the averaged density matrix and derive its explicit form perturbatively for a suitable class of couplings. The result naturally separates into a quadratic exponential term, which governs the suppression of phase sensitive contributions in the averaged description, and a purely oscillatory contribution, which modifies coherent propagation through a disorder-induced correction to the propagation kernel. This provides a unified description of interference, diffraction, and statistical fluctuations of the lens distribution within a single framework. We also identify the physical scales controlling the onset of coherence loss and illustrate the formalism in the case of Gaussian wave packets. More generally

Source URL. <http://arxiv.org/abs/2604.15313v1>

31

AnimationBench: Are Video Models Good at Character-Centric Animation?

arXiv

2026-04-16

Abstract. Video generation has advanced rapidly, with recent methods producing increasingly convincing animated results. However, existing benchmarks—largely designed for realistic videos—struggle to evaluate animation-style generation with its stylized appearance, exaggerated motion, and character-centric consistency. Moreover, they also rely on fixed prompt sets and rigid pipelines, offering limited flexibility for open-domain content and custom evaluation needs. To address this gap, we introduce AnimationBench, the first systematic benchmark for evaluating animation image-to-video generation. AnimationBench operationalizes the Twelve Basic Principles of Animation and IP Preservation into measurable evaluation dimensions, together with Broader Quality Dimensions including semantic consistency, motion rationality, and camera motion consistency. The benchmark supports both a standardized close-set evaluation for reproducible comparison and a flexible open-set evaluation for diagnostic analysis, and leverages visual-language models for scalable assessment. Extensive experiments show that AnimationBench aligns well with human judgment and exposes animation-specific quality differences overlooked

Source URL. <http://arxiv.org/abs/2604.15299v1>

32

Reed--Muller Codes Achieve the Symmetric Capacity on Finite-State Channels

arXiv

2026-04-16

Abstract. We study reliable communication over finite-state channels (FSCs) using Reed--Muller (RM) codes. Building on recent symmetry-based analyses for memoryless channels, we show that a sequence of binary RM codes (with some random scrambling) can achieve the symmetric capacity (or uniform-input information rate) of a binary-input indecomposable FSC. Our approach has three components. First, we establish a capacity-via-symmetry theorem for doubly-transitive group codes on discrete memoryless channels (DMCs) with non-binary inputs, under some symmetry and puncturing conditions. Then, we reduce a binary-input FSC to an almost memoryless non-binary channel by grouping adjacent input bits into blocks and interleaving non-binary codes onto the channel. Finally, we show that the interleaved non-binary codes can be constructed from a single binary RM code.

Source URL. <http://arxiv.org/abs/2604.15295v1>

33

Borel--Bernstein and Hirst-type Theorems for Nearest-Integer Complex Continued Fractions over Euclidean Imaginary Quadratic Fields

arXiv

2026-04-16

Abstract. For each $d \in \{1, 2, 3, 7, 11\}$, let T_d be the nearest-integer complex continued fraction map associated with the Euclidean ring $\mathcal{O}_d$, and let (a_n) be its digit sequence. We prove two metric results for this five-system family. First, for every sequence $(u_n)_{n \geq 1}$ with $u_n \geq 1$, the set of points for which $|a_n| \geq u_n$ for infinitely many n has full or zero normalized Lebesgue measure according as $\sum_{n=1}^{\infty} u_n^{-2}$ diverges or converges. This gives a unified Borel--Bernstein theorem, extending the Hurwitz case $d=1$ to all five Euclidean imaginary quadratic fields. Second, for any infinite set $S \subset \mathcal{O}_d$, if $\tau(S)$ denotes its convergence exponent, then the digit-restricted set $F_d(S) = \{z : a_n(z) \in S \text{ for all } n, |a_n(z)| \rightarrow \infty\}$ satisfies $\dim_H F_d(S) = \tau(S)/2$. More generally, for any cutoff function $f(n) \rightarrow \infty$, the set $F_d(S, f) = \{z \in F_d(S) : |a_n(z)| \leq f(n) \text{ for all } n\}$ is either empty or has the same Hausdorff dimension $\tau(S)/2$. The proof combines quantitative ergodic properties of the nearest-integer systems with a large-digit conformal iterated function subsystem that is 2 -decaying. We also

Source URL. <http://arxiv.org/abs/2604.15293v1>

34

Cloudy with a chance of metals: Indications of CO₂ in the atmosphere of GJ 1214 b from high-resolution K-band spectroscopy

arXiv

2026-04-16

Abstract. Sub-Neptune exoplanets frequently exhibit muted transmission spectra, with GJ 1214 b being the most prominent example. Following years of intense observing campaigns yielding featureless planetary spectra, recent observations with JWST revealed the first possible atmospheric signatures. We present high-resolution transmission spectroscopy of GJ 1214 b based on eight transits obtained with the CRIRES+ spectrograph in the K band. We used SYSREM to remove telluric and stellar signals and searched for signatures of H₂O, CO, CH₄, H₂S, NH₃, and CO₂ using the cross-correlation technique. We obtained non-detections for the first five molecules and used injection recovery tests to derive upper limits on the atmosphere. For CO₂ we measure a CCF signal at $S/N \sim 3.6$, with a detailed investigation showing no obvious indication that it is caused by correlated noise. A Welch t-test confirmed the in-trail and out-of-trail distributions to be different at 3.4σ confidence. A Bayesian retrieval framework with free chemistry, resulted in volume mixing ratios corresponding to a metallicity of $[M/H] = 0.48^{+0.89}_{-1.70}$, an opacity deck pressure of $\log_{10}(P_{\text{c}}) = -3.04^{+}$

Source URL. <http://arxiv.org/abs/2604.15292v1>

Abstract. The biological sensorimotor system is a source of inspiration for the design of neuromorphic ballistic control systems. A large portion of sensorimotor-inspired research focuses on the sensory encoding and information processing stages of the system. However, research on broader task-performance systems, involving actuator control on the output side remains scarce. In this work, we develop and train a neuromuscular-inspired model to perform ballistic control. In the model, a spiking neural network's output spikes are used to generate twitch-like signals. These twitches are the basis for generating a continuous fluctuating output signal that is used to operate an actuator. We refer to the used model as the Twitch Neural Network (TwNN). As a test case, the model is trained to control the paddle of an adapted version of the game of Pong. An adapted version of the Direct Feedback Alignment learning rule, specifically for integrate-and-fire neurons, is introduced. The new rule avoids the update-locking problem of backpropagation, allowing network weight updates in parallel. The model output consists of one group of agonist-innervating motor neurons, and one group of antagonist-inner

Source URL. <https://openalex.org/W7154134978>

Abstract. Soft robotic grippers excel in unstructured manipulation but suffer catastrophic failure rates (72%) when grasping deformable organics, fabrics, and mixed debris due to hyperchaotic pneumatic dynamics. This paper introduces the first Lyapunov stability controller for soft robotics, deploying real-time maximal Lyapunov exponent estimation (λ_{MLE}) from fibre-optic strain sensor arrays running at 100Hz on Intel Loihi 2 neuromorphic chips. The system reconstructs 12D phase space embeddings via Takens theorem, detecting chaos onset 187ms early during dual-material transitions (tomato $\rightarrow$ bolt), enabling pre-emptive damping that transforms strange attractors into stable limit cycles. Experimental validation across USDA organic datasets (tomatoes, grapes, leafy greens) and MRF waste streams demonstrates 94.2% grasp success 3.7 $\times$ improvement over PID baselines with 2.3 $\times$ faster cycles (2.1 grips/second) and 67% energy savings. Neuromorphic acceleration achieves 187 μ s latency for 12D divergence computation, 28 $\times$ faster than GPU methods. Field deployments confirm robustness, agricultural harvesting sustains 3 clusters/minute, waste sorting handles mixed-material chaos, and medical tissue manipula

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